

RESEARCH & INNOVATION

Social Media-Based Credit Card Recommendation System

Leveraging Behavioral Signals for Inclusive & Ethical Financial Recommendations

Problem Statement

1

Excluded Populations

Students, gig workers & young professionals lack formal credit history, making them ineligible under traditional models.

2

Unstructured Data Gap

Social media contains rich behavioral signals about trust & affluence, but no reliable framework exists to extract them.

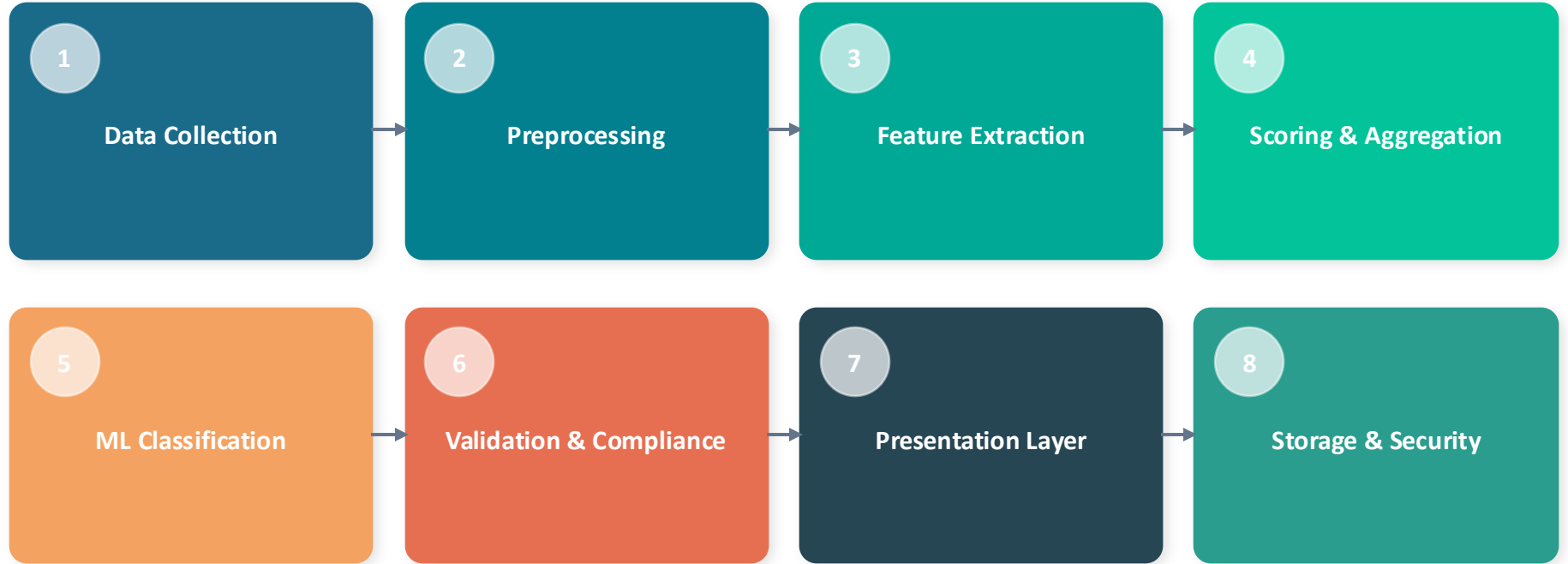
3

Bias & Privacy Risks

Naive use of social data introduces bias, privacy violations, and fairness concerns without proper safeguards.

Objective: Extract trust & affluence signals from consented social media data to recommend appropriate credit card categories — ensuring fairness, privacy, and explainability.

System Architecture Overview



Feature Extraction: Affluence & Trust Signals

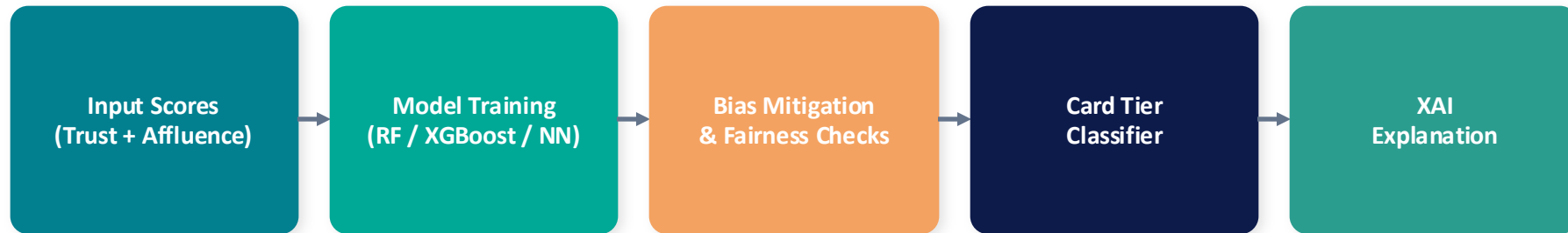
Affluence Signal Extractor

- NLP brand & luxury keyword detection
- CV image classification (luxury items, travel, dining)
- Location & check-in frequency analysis
- Purchase & lifestyle mention extraction

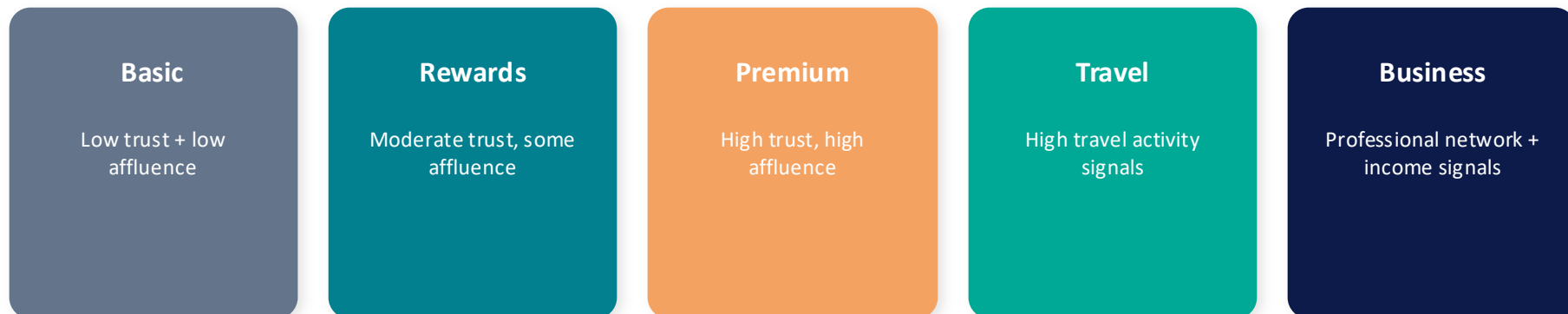
Trust Signal Extractor

- Account age & activity consistency analysis
- Network credibility scoring (verified connections)
- Sentiment & tone consistency analysis
- Spam & bot detection filters

ML Pipeline & Credit Card Tier Mapping



Credit Card Tier Output



Bias Mitigation, Privacy & Compliance



Fairness & Bias

- Demographic parity checks across groups
- Fairness-aware weighting adjustments
- Periodic bias audits on model output
- Human-in-the-loop for borderline cases



Privacy & Security

- OAuth-based user consent & auth
- PII masking before processing
- Encrypted storage (at rest & in transit)
- Auto data deletion post-processing



Compliance

- GDPR & RBI data protection compliance
- Audit logging for all model decisions
- Data retention policy engine
- SHAP/LIME-based XAI explanations

Conclusion & Future Roadmap

✅ Social signals can bridge the credit history gap for underserved populations.

🧠 Trust + Affluence scoring with ML enables nuanced, tiered card recommendations.

⚖️ Bias mitigation and compliance modules ensure ethical, fair outcomes.

🔒 Consent-first design and PII masking protect user privacy throughout the pipeline.

Future Roadmap

- Real-time API integration
- Federated learning for privacy
- Multi-modal behavioral scoring
- Cross-institution data sharing (with consent)
- RegTech integration for auto-compliance